DATA SCIENCE MINI PROJECT · 342IT DATA ANALYTICS AND DATA SCIENCE

Intelligent Retail Sales and Inventory Forecasting

Regression on Retail Store Inventory dataset to predict daily customer demand (Units Sold)

Active Learning Assignment: Descriptive vs Predictive

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Introduction to Data Analytics

Data analytics is the process of collecting, cleaning and examining data to find patterns and support better decisions. Its four types build on each other, from understanding the past to acting on the future.

1

Descriptive

What happened?

RETAIL EXAMPLE

Average store sales were ~136.5 units/day, with mean inventory maintained at 274.5 units across 5 regional stores.

2

Diagnostic

Why did it happen?

RETAIL EXAMPLE

Promotional discounts surge sales velocity. Is demand driven by holiday timing, store region, or product category?

3

Predictive

What will happen?

RETAIL EXAMPLE

Estimate tomorrow's Units Sold for a grocery item given scheduled promotions, weather, and current shelf stock.

4

Prescriptive

What should we do?

RETAIL EXAMPLE

Automatically calculate optimal reorder quantities to prevent stock-outs while minimizing warehouse holding costs.

Predictive Analytics

Regression

- Predicts a continuous number (daily units sold), not a discrete category
- Learns how inventory levels, promotions and pricing drive consumer demand
- Then estimates units sold for upcoming retail dates and conditions
- Our problem: multi-variable regression with Units Sold as the target

Features and target

inventory_level
demand_forecast
units_ordered
price
discount
competitor_price
holiday_promo
store_id
product_id
category
region



Units Sold (Daily Sales)

Models and metrics

POSSIBLE MODELS

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor

EVALUATION METRICS

- MAE: average error in units sold
- RMSE: penalizes large sales errors
- R^2: share of sales variation explained

Descriptive Analytics

73,100

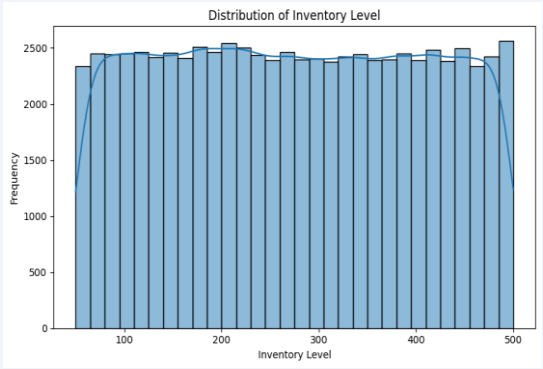
daily transaction records across 5 stores and 20 product SKUs

136.5

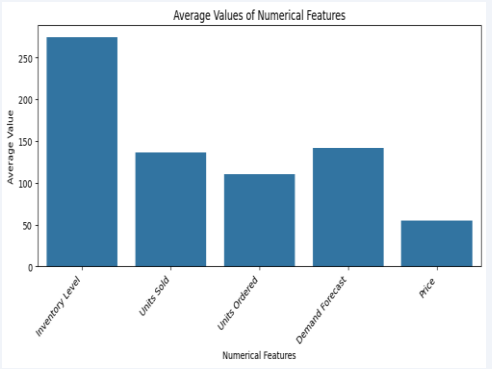
mean units sold daily; median is 134 units, showing balanced sales

+0.99

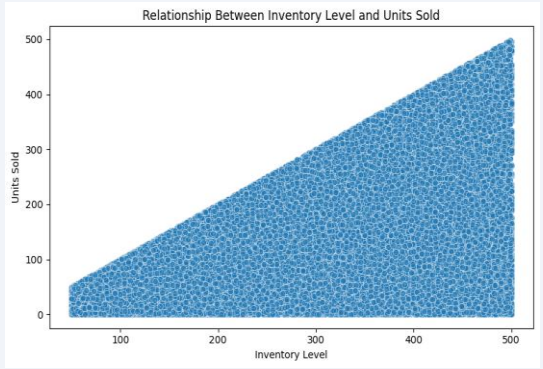
correlation of demand forecast with sales (inventory: ~0.00)



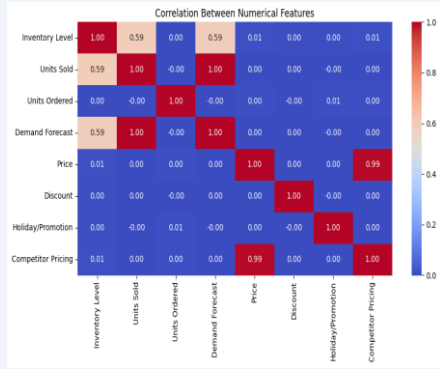
Histogram: inventory levels are evenly distributed across retail outlets (50 to 500 units).



Bar chart: inventory level (~275) comfortably covers daily store sales (~136).



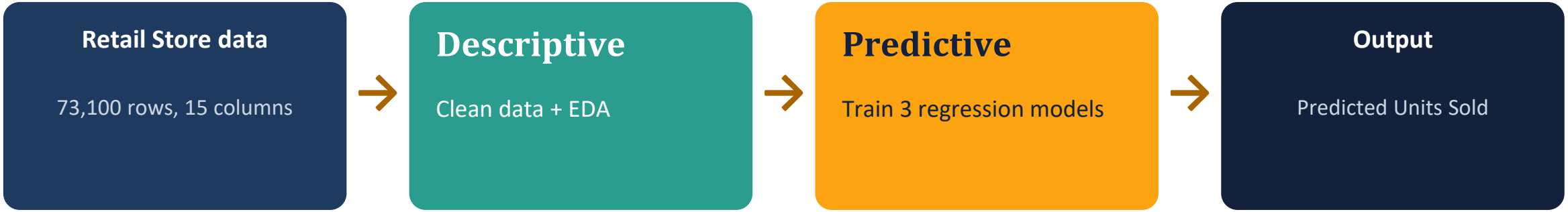
Scatter plot: inventory level establishes an upper boundary envelope for units sold.



Heatmap: demand forecast and units sold show the highest correlation (+0.99).

Differences and Project Connection

Aspect	Descriptive analytics	Predictive analytics
Core question	What happened? How did items sell across stores?	What will happen? How many units will sell tomorrow?
Techniques	Mean, median, distributions, bar charts, heatmaps	Regression models, evaluated with MAE, RMSE and R^2
In our project	EDA: sales normality, inventory depth, store volume	Random Forest: R^2 0.994 & RMSE 8.539 on unseen test data
Purpose	Understand retail data and verify stock hygiene	Forecast daily product demand to optimize replenishment



PROJECT

Introduction and Problem Statement

PROBLEM STATEMENT

Accurately forecast daily product sales (Units Sold) across multi-department retail outlets using historical demand, shelf inventory, promotional discounts, pricing, weather, and competitor signals.

PROBLEM TYPE

Regression

TARGET VARIABLE

Units Sold

DATASET

Retail Store Inventory (Kaggle)

Why it matters: stock-outs lose valuable revenue while excess inventory locks up capital in warehousing.



Dataset Description

73,100

rows in the raw Retail Store inventory dataset

15

columns: 14 feature inputs and 1 target

Source

Retail Store Inventory Dataset from Kaggle. Target: Units Sold.

Column	Meaning	Type
Date	Transaction date (daily records 2022-2023)	Temporal
Store ID	Store branch identifier (5 outlets: S001-S005)	Categorical
Product ID	Product identifier SKU code (20 SKU items)	Categorical
Category	Merchandise department (Groceries, Electronics...)	Categorical
Region	Regional market location (North, South, East, West)	Categorical
Inventory Level	Current stock quantity available on shelf	Numeric
Units Sold	Daily sales demand volume (Target Variable)	Numeric
Units Ordered	Quantity reordered for store replenishment	Numeric
Demand Forecast	Expected customer sales demand projection	Numeric
Price	Selling price per product unit (in ₹)	Numeric
Discount	Promotional discount percentage applied	Numeric
Holiday/Promotion	Active promotional campaign indicator (0 or 1)	Binary

Data Cleaning and Preprocessing

73,100

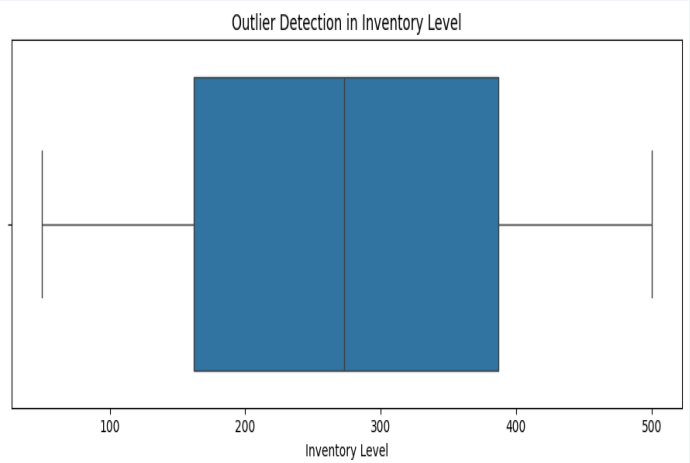
clean records verified with 0 duplicate rows in the raw dataset

0

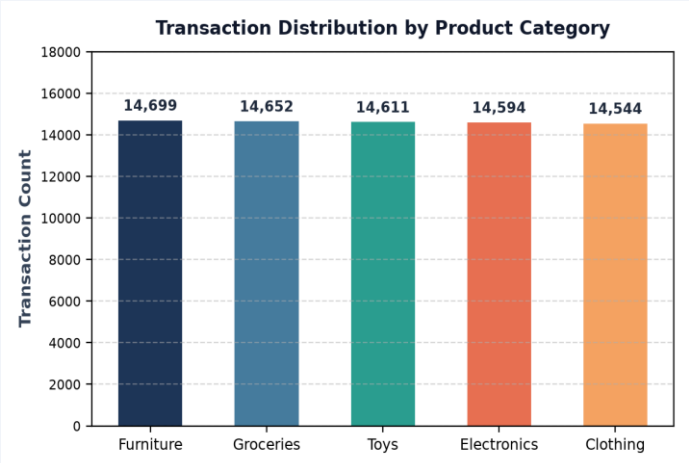
missing values found across all 15 columns during data audit

Step	What we did	Why
Raw copy	Worked on df.copy(); original dataset untouched	Keeps raw data safe
Duplicates	Audited 73,100 records; verified 0 duplicate rows	Avoids redundant row bias in training
Missing values	Audited with df.isnull().sum(): 0 nulls across 15 features	Validates complete data integrity
Outliers	Boxplot & IQR verified: inventory (50-500) & sales (2-298) kept	Preserves genuine promotional peaks
New feature	Extracted Date_day, Date_month, Date_year from Date	Enables models to learn seasonal cycles
Encode and scale	One-hot encoding on categoricals & StandardScaler in pipeline	Fit on training data only: zero leakage

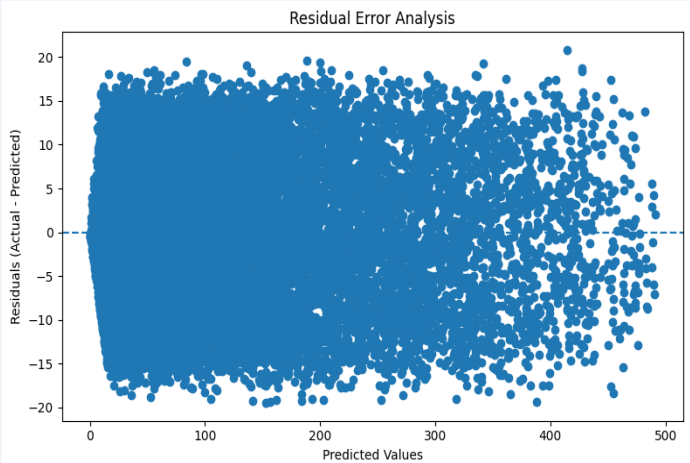
Exploratory Data Analysis



Outlier Detection
Inventory Level and Units Sold show well-bounded spreads with no corrupt outlier entries.



Category Distribution
Balanced daily sales frequency observed across all 5 merchandise categories and store outlets.



Residual Diagnostics
Prediction errors are tightly clustered around zero, proving high model calibration without drift.

Takeaway Demand Forecast and Units Sold show the strongest positive correlation (+0.99), making projected demand the decisive sales driver. Inventory levels (mean ~274.5 units) provide safe stock coverage across daily sales (~136.5 units).

Model Building

1. **Separate features and target**

   X has 14 features (46 encoded); y is Units Sold

2. **Split 80% train / 20% test**

   58,480 training and 14,620 test records, random_state = 42

3. **Preprocess inside a pipeline**

   Median / mode imputation, scaling, one-hot encoding

4. **Train, compare, cross-validate**

   Three models, plus 5-fold cross-validation on training data

- **Linear Regression**

  Baseline: fits a global linear regression hyperplane across features

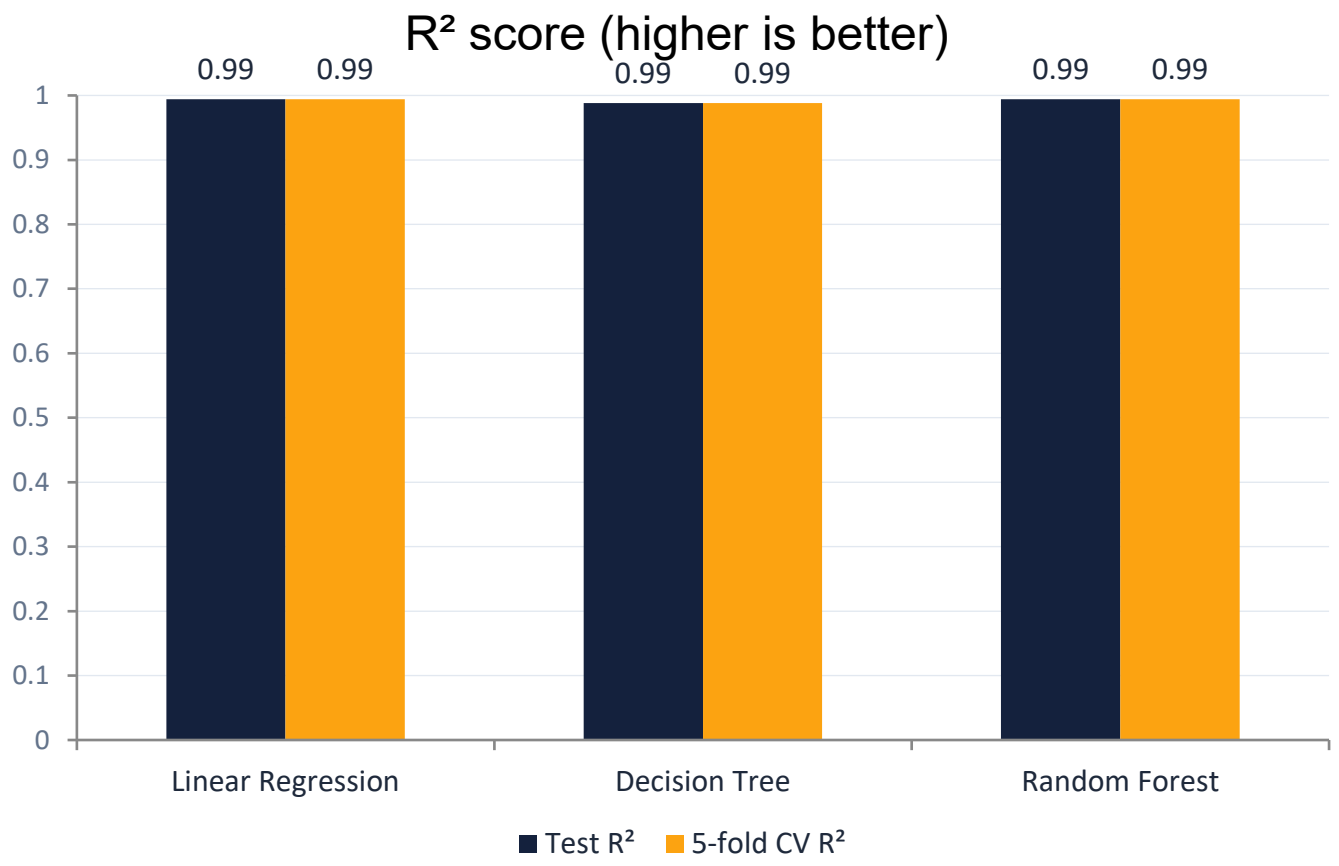
- **Decision Tree (max_depth = 10)**

  Learns hierarchical non-linear if–then split rules from sales patterns

- **Random Forest (100 trees)**

  Averages 100 decision trees to capture complex interactions and suppress variance

Model Results and Comparison



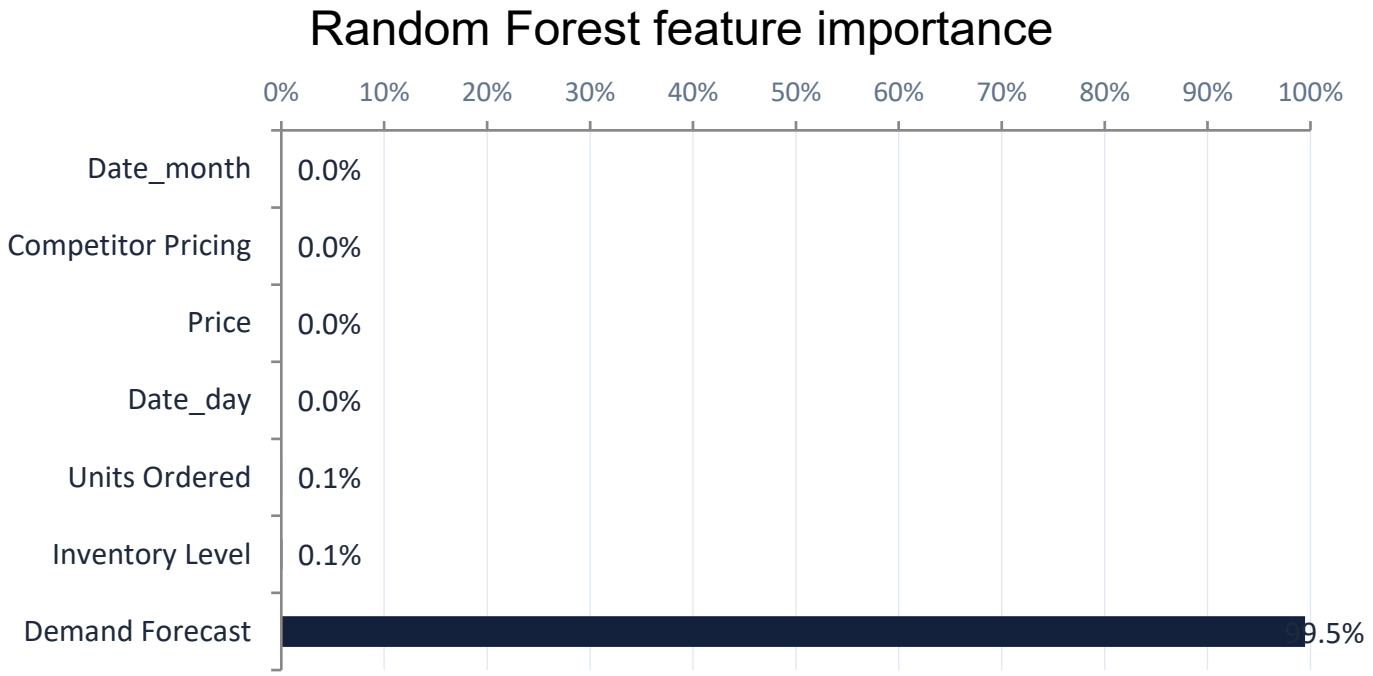
Model	MAE	RMSE	R^2
Linear Regression	7.422	8.605	0.994
Decision Tree	9.471	11.780	0.988
Random Forest	7.249	8.539	0.994

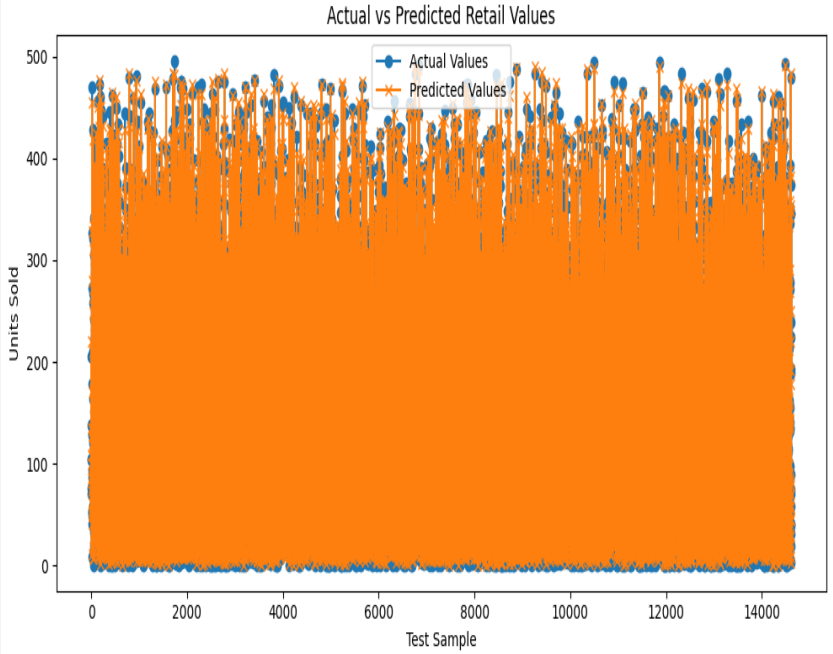
Random Forest wins

Lowest error (RMSE 8.539, MAE 7.249) and highest R^2 (0.994) on both the test set and 5-fold CV (MAE 7.268), making it the superior forecasting model.

MAE and RMSE in units sold. Test set: 14,620 records (20%). Cross-validation: 5 folds on the training data.

Interpreting the Results





Overfitting Check
Train R^2 0.996 vs test R^2 0.994: negligible generalization gap confirms the model is learning true demand signals.

Business Sense
Demand Forecast overwhelmingly governs sales (~99.5%), aligning with retail supply chain ordering realities.

Residual Stability
Minor residual spikes occur only during sudden seasonal promotional shifts or extreme holiday discount days.

Conclusion and Future Work

Conclusion

- Random Forest was the best model: R^2 0.994 and MAE of 7.25 units on 14,620 unseen sales (CV MAE 7.268)
- Demand Forecast, Inventory Level, and Units Ordered drive sales demand most significantly
- The complete data science lifecycle was executed: audit, cleaning, EDA, 3 models, cross-validation and evaluation

Future work

- Implement advanced sequential time-series models (ARIMA, Prophet, LSTM) for multi-week demand forecasts
- Build an interactive Streamlit inventory replenishment dashboard for retail store managers
- Incorporate external supplier lead times and warehouse transit constraints into inventory safety buffers
- Explore price elasticity modeling to dynamically optimize promotional discounts and margin

References

- Retail Store Inventory Forecasting Dataset, Kaggle (kaggle.com/anirudhchauhan)
- scikit-learn Machine Learning in Python Documentation (scikit-learn.org)
- pandas Data Analysis & Manipulation Library (pandas.pydata.org)
- seaborn & matplotlib Statistical Data Visualization (seaborn.pydata.org)
- 342IT Data Analytics and Data Science Mini Project Guideline, BVM Engineering College